ARITHMETIC, AGE AND COMPARISON PROBLEMS

Cutting Problems, Ages and Ordering by Weight

OAVS TGT | Reasoning (Additional topic from the “etc.” in the syllabus) | Topic 26

Syllabus line (Reasoning, 30 Marks): The verbal and non-verbal topics listed in the syllabus end with “etc.”. This topic is not named in that line, but it is asked in the papers: arithmetic word problems appear in English 2019 (Q32 cutting, Q41 age) and Social Studies 2023 (Q23 weights).

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Three questions appear (English 2019 Q32, Q41 and Social Studies 2023 Q23). All three are solved below. The practice questions were generated or checked by program so that each has exactly one correct option.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The 3 PYQs and their types
2	Foundation	Cutting rule, age equations, comparison chains
3	Solved PYQs	3 PYQs with step-by-step reasoning
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Three questions, two papers

Paper	Q.	Question	Type	Ans
English 2019	32	Rolls cut into 10 pieces each; 36 cuts a minute; how many rolls in 26 minutes?	Cuts and pieces	A (104)
English 2019	41	Vamshi, 16, was 4 times as old as Raju; his age when twice Raju's age	Age	A (24)
Social Studies 2023	23	Six rods P, Q, R, S, T, U with weight relations; which is heaviest?	Comparison	2 (R)
English 2022 / Mathematics 2024 / Science 2024	–	none		

PYQ INSIGHT: Three different word problems, each needing **one equation or one chain**. Cutting problems turn on "a straight cut gives one more piece", age problems turn on "both people age by the same number of years", and comparison problems turn on **writing everything in terms of one weight**.

Pattern	PYQs	Priority
Cutting problems (cuts and pieces)	1	★★★ Must
Age problems (a stated age and a ratio)	1	★★★ Must
Comparison / ordering by weight or size	1	★★★ Must
Ratio and sum problems	0	★★ Expected

CHAPTER 2

Foundation

Cutting rule, age equations and comparison chains

2.1 Cutting problems

Situation	Rule
Cut a rod/roll into n pieces with straight cuts in one line	Number of cuts = n − 1
Total cuts available	Rolls = total cuts ÷ cuts per roll
Time per cut given	Total time = cuts × time per cut

2.2 Age problems

Idea	How to use it
Both people age equally	After t years, add t to both ages; t years ago, subtract t from both
Ratio given	Write ages as ratio parts (5x, 3x) and use the extra condition to find x
“Times as old”	Write an equation: older age = k × younger age at that time
Age difference	The difference never changes

SHORTCUT: English 2019 Q41: Vamshi is 16 and 4 times Raju, so Raju is 4. After t years: 16 + t = 2(4 + t), so t = 8 and Vamshi is 24. Using the difference: difference = 12; when Vamshi is twice Raju, Raju = 12, Vamshi = 24.

2.3 Comparison (weights, heights, ages)

Step	Do this
1	Choose one object as the base and name its weight x (or use a given weight).
2	Write each other weight in terms of x from the statements.
3	Compare the expressions or substitute a number.
4	Keep track of “twice”, “half” and “less than” carefully.

CHAPTER 3

Solved PYQs

All three questions with step-by-step reasoning

English 2019 Q32

An artist cuts canvas pieces from a roll. He cuts each roll of equal length into 10 pieces. He makes 36 cuts in a minute. How many rolls would be cut in 26 minutes? (A) 104 (B) 26 (C) 52 (D) 35

1. Cutting a roll into 10 pieces needs $10 - 1 = 9$ cuts.
2. In 26 minutes he makes $26 \times 36 = 936$ cuts.
3. Number of rolls $= 936 \div 9 = 104$.
Answer: Option A (104)

English 2019 Q41

16-year-old Vamshi was 4 times as old as his brother Raju. How old will Vamshi be when he is twice as old as Raju? (A) 24 (B) 12 (C) 8 (D) 16

1. Raju's age $= 16 \div 4 = 4$ years.
2. After t years: $16 + t = 2 \times (4 + t) = 8 + 2t$, so $t = 8$.
3. Vamshi's age then $= 16 + 8 = 24$.
Answer: Option A (24)

Social Studies 2023 Q23

Among six iron rods P, Q, R, S, T and U, each of different weight: P is half of Q; R is twice S; T is 5 kg less than R; U is some kg less than S; Q is half of S. Which is the heaviest rod? (1) S (2) R (3) Q (4) T

1. Let $S = 2k$. Then $Q = S \div 2 = k$, $P = Q \div 2 = k/2$.
2. $R = 2 \times S = 4k$, so R is greater than S.
3. $T = R - 5$, so $T < R$. U is less than S, so $U < S < R$. Q (= k) and P (= k/2) are smaller than S.
4. So R is heavier than every other rod. The heaviest rod is R.
Answer: Option 2 (R)

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Pieces and cuts:** cuts = pieces − 1 (one line).
- **Equal ageing:** add or subtract the same years from both.
- **Base variable:** express every quantity in terms of one unknown.
- **Check with numbers:** put a value in the base and compute all.
- **Time:** 45 seconds per question.

Trap	How to avoid it
Using pieces as cuts	10 pieces need 9 cuts.
Adding years to only one person	Both people age equally.
Misreading “half” and “twice”	Write the equation before solving.
Comparing without a base	Express everything in one variable.

Scope

Related PYQs / topics	Where
Ranking in a queue or class	Ranks (earlier topic)
Ratio and percentage questions	Outside the reasoning syllabus

CHAPTER 5

Practice Set

6 original questions; answers on the next page

A. Cutting problems

Q1. A tailor cuts each roll of cloth into 8 pieces. He makes 28 cuts every minute. How many rolls does he cut in 15 minutes?

A. 52 B. 60 C. 56 D. 70

Q2. A 60 m rope is cut into pieces of 5 m each. Each cut takes 2 minutes. How long does it take to cut the whole rope?

A. 24 minutes B. 20 minutes C. 26 minutes D. 22 minutes

B. Age problems

Q3. A father is 5 times as old as his son. After 6 years he will be 3 times as old as his son. What is the son’s present age?

A. 6 years B. 8 years C. 4 years D. 10 years

Q4. The ages of A and B are in the ratio 5 : 3 and their sum is 48 years. What will be A’s age after 6 years?

A. 30 years B. 34 years C. 36 years D. 40 years

Q5. Seema is 3 times as old as her daughter. 10 years ago she was 7 times as old as her daughter. What is Seema’s present age?

A. 40 years B. 45 years C. 36 years D. 50 years

C. Comparison

Q6. P weighs 20 kg. Q weighs twice P. R weighs three times P. S weighs 10 kg less than R. T weighs half of S. U weighs 5 kg more than Q. Which is the lightest?

A. T B. Q C. P D. U

Answer key

Q	Answer	Reason
1	B	Each roll needs 8 − 1 = 7 cuts. In 15 minutes he makes 28 × 15 = 420 cuts. 420 ÷ 7 = 60 rolls.
2	D	60 ÷ 5 = 12 pieces need 11 cuts. Time = 11 × 2 = 22 minutes.
3	A	Let the son be x: 5x + 6 = 3(x + 6) gives 2x = 12, x = 6.
4	C	5x + 3x = 48 gives x = 6. A = 30. After 6 years, A is 36.
5	B	Let the daughter be x: 3x − 10 = 7(x − 10) gives 4x = 60, x = 15. Seema = 45.
6	C	P = 20, Q = 40, R = 60, S = 50, T = 25, U = 45. The lightest is P (20 kg).

CHAPTER 6

Quick Revision

One page for the last day

Type	Rule
Cuts	Cuts = pieces − 1
Age	Add or subtract the same years from both; use the difference
Ratio	Use parts 5x, 3x and find x
Comparison	Base variable; substitute a number

PYQ RULE: English 2019 Q32: cuts = pieces − 1; Q41: equal ageing; Social Studies 2023 Q23: chain of weights in one variable.

ROUTINE: Exam-day routine: write the unknown; build the equation; solve; check. 45 seconds per question.